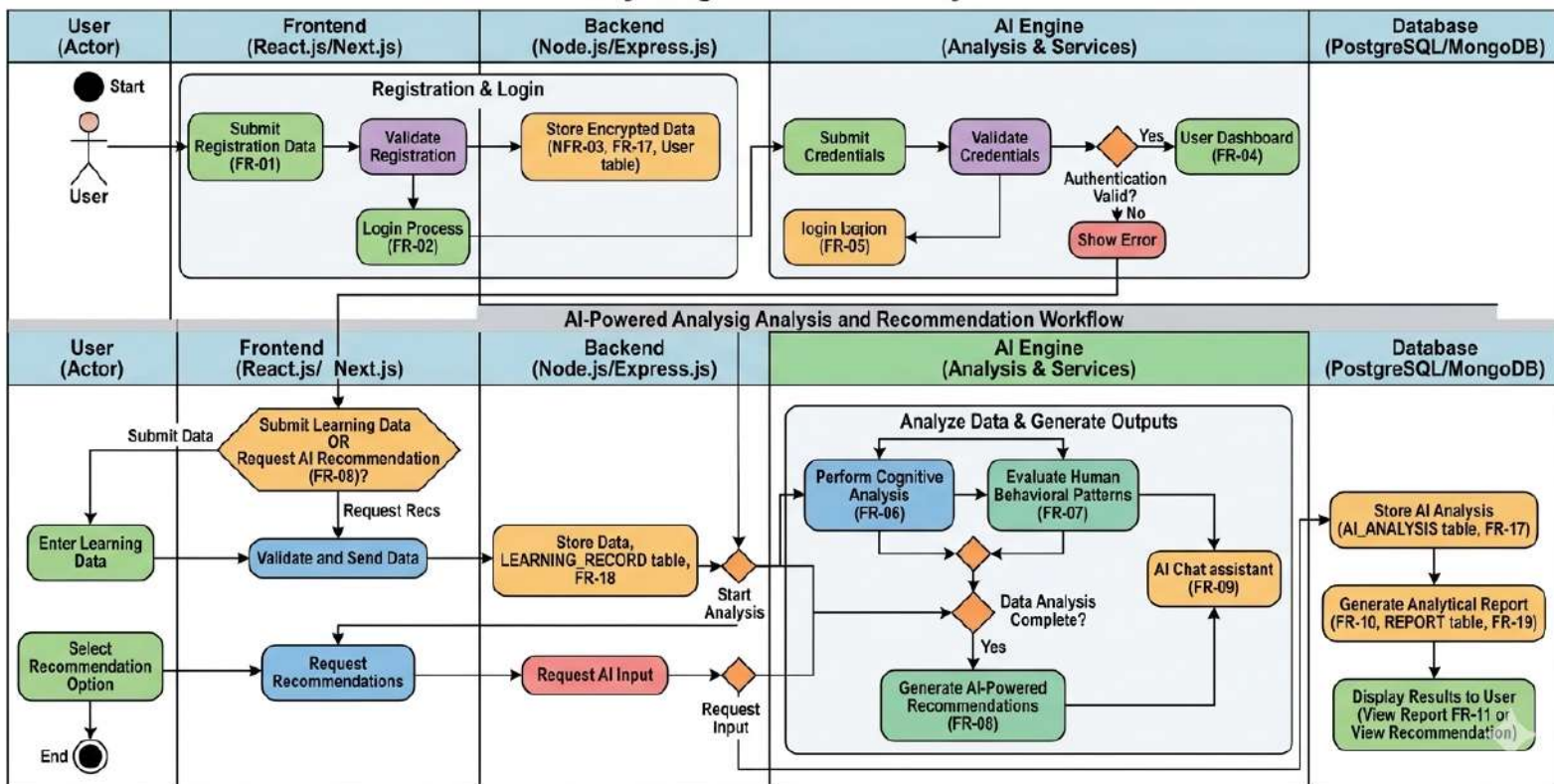
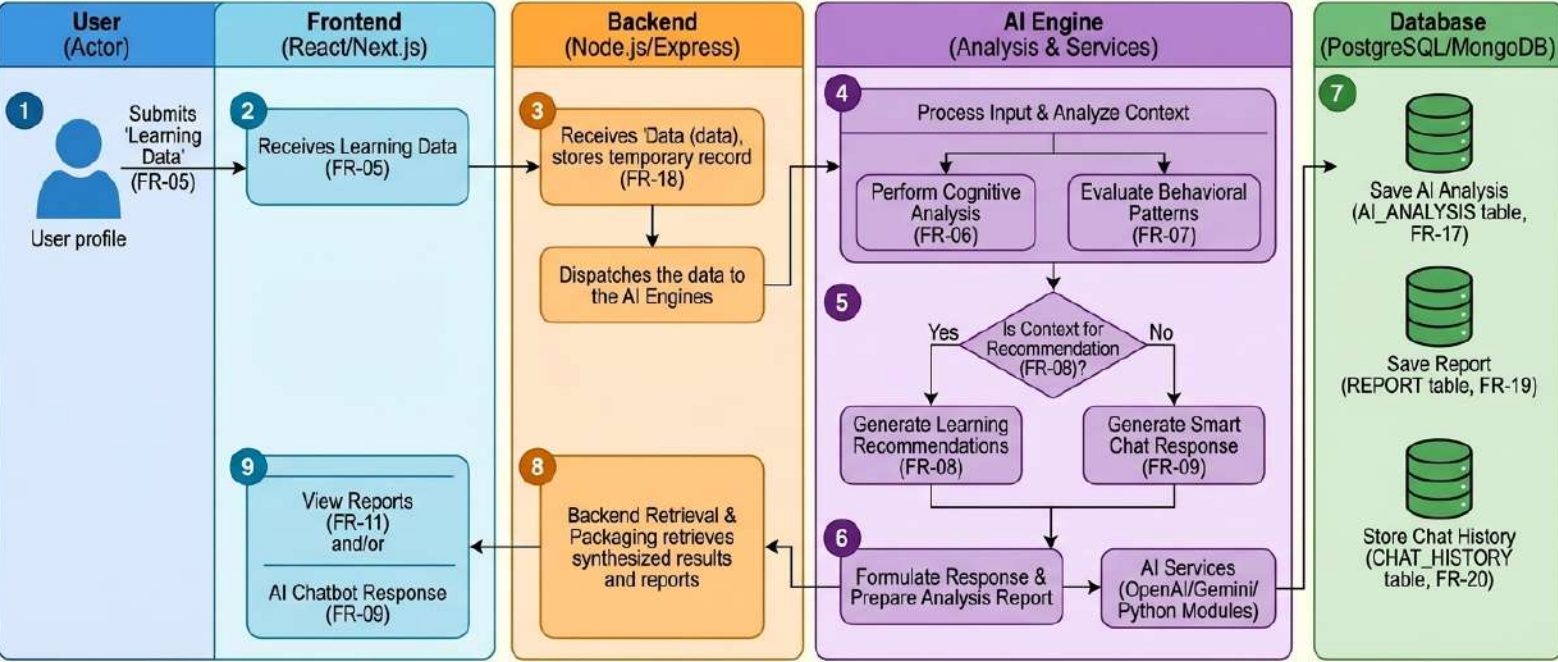


Chapter 4
Activity Diagrams for NeuroSync AI



Chapter 4

AI Integration Workflow for NeuroSync AI platform



- Why AI is needed (Mandatory):**

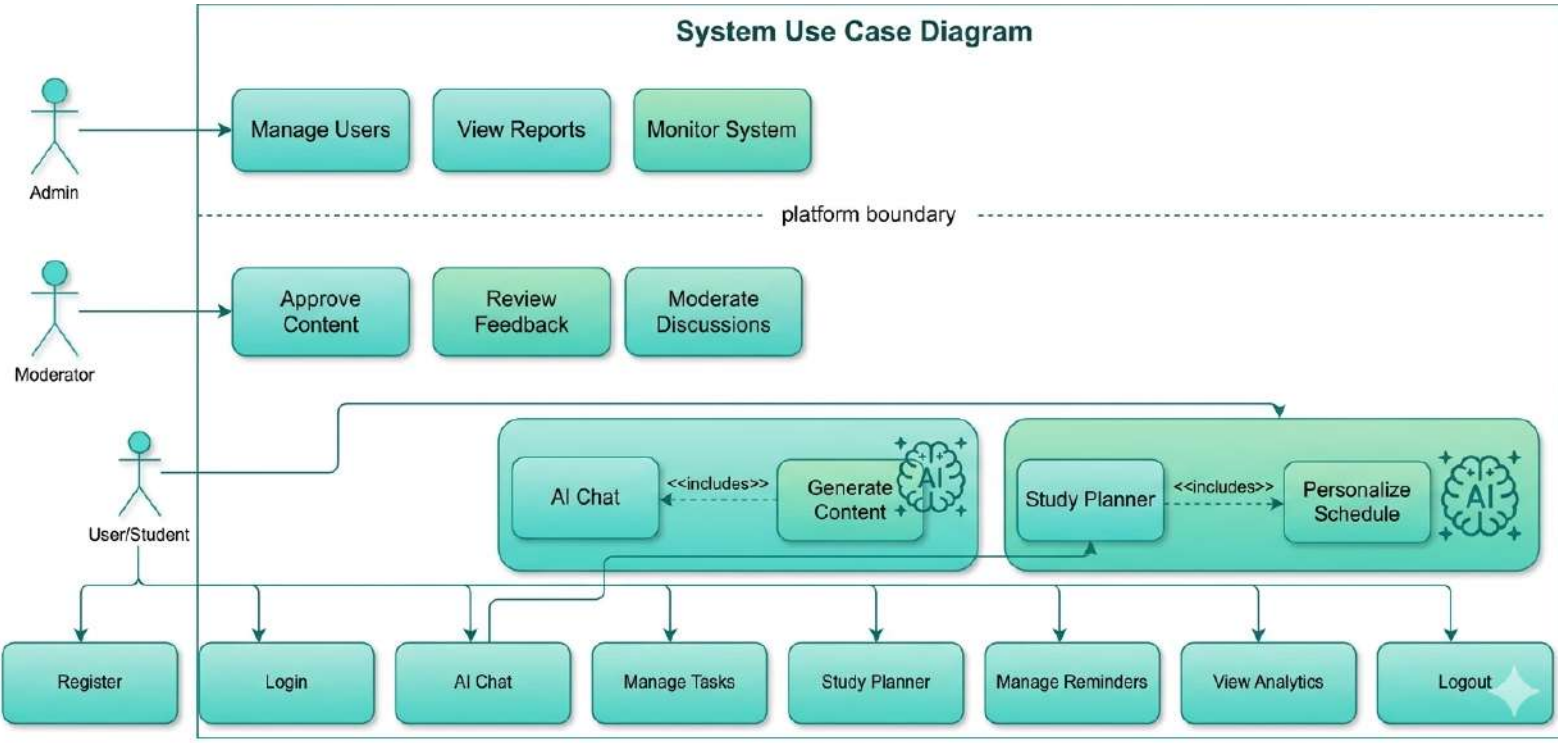
 - Personalize learning experience
 - Identify individual cognitive styles
 - Detect behavioral engagement gaps
 - Analyze complex learning data in real-time
- AI Services/Models Used (Mandatory):**

 - OpenAI API (GPT for Chat/Language)
 - Gemini API (Context/Language)
 - Hugging Face (Embeddings/Specific Tasks)
 - Python AI Modules (Specific cognitive/behavioral algorithms)
- Input to AI System (Mandatory):**
Learning Data (FR-05), user behavior patterns, query context

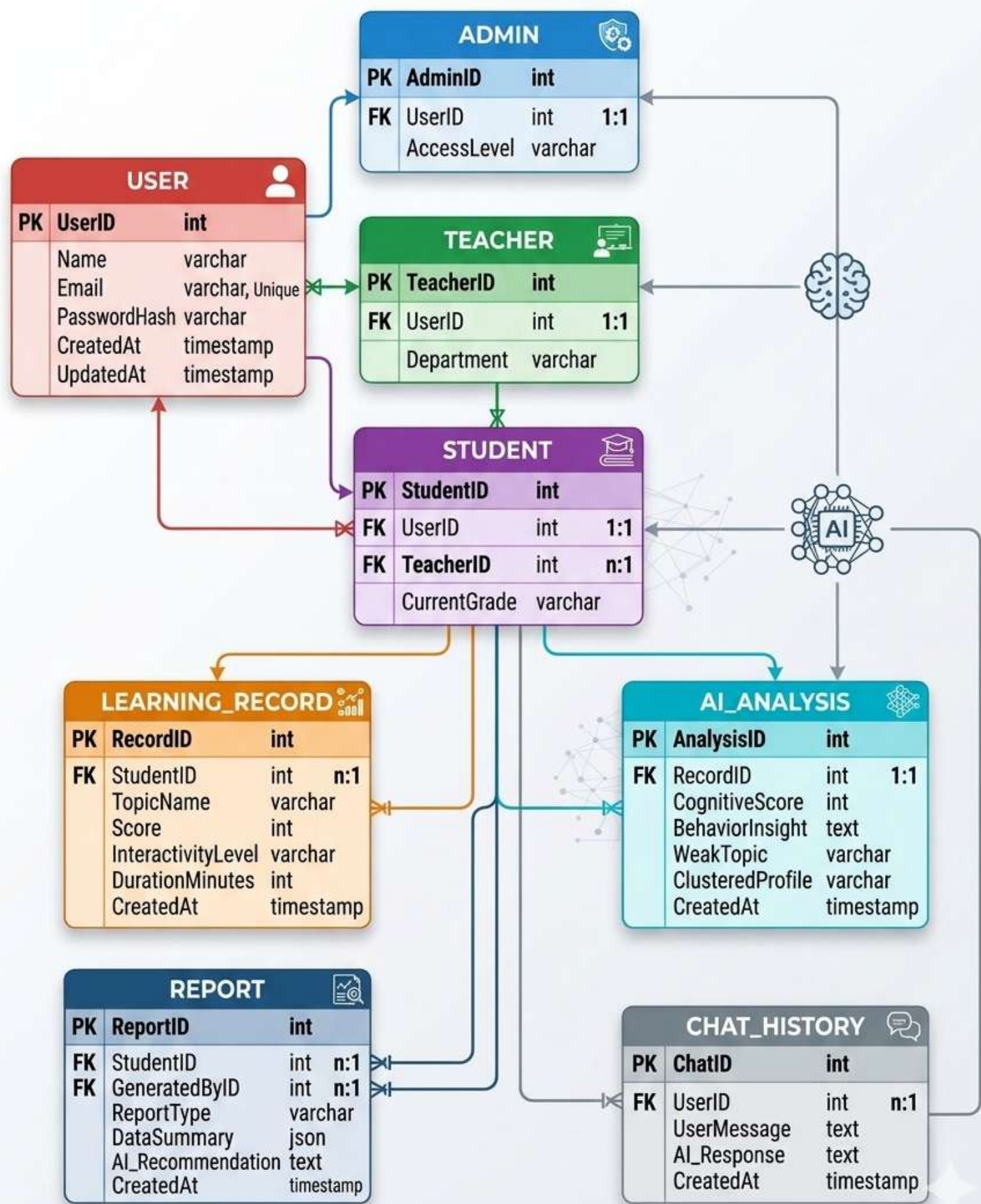
Output from AI System (Mandatory):
Cognitive Analysis (FR-06), Behavior Assessment (FR-07), Learning Recommendations (FR-08), Smart Chat Responses (FR-09)
- How AI Improves Project (Mandatory):**

 - Enables personalized learning (Objective 1.4)
 - Solves adaptive learning problem (Problem Statement 1.3)
 - Automates analysis for reporting (Objectives 1.4, FR-10)
 - Improves student engagement and performance

System Use Case Diagram



NeuroSync AI: ER Diagram (PostgreSQL)



SYSTEM ARCHITECTURE



SYSTEM API ARCHITECTURE DIAGRAM

